DATA ANALYST INTERNSHIP

WEEK 4 – CAPSTONE PROJECT

PROJECT TITLE	Sales Data Analysis and Prediction
SUBMITTED BY	Janhavi Amol Chavan
INTERNSHIP	Data Analyst Internship
TOOLS USED	Python, Excel and Power BI
PROJECT DATASET	Sample Sales Dataset

A practical study of sales performance, business patterns and predictive modelling using a sample business sales dataset.

Project Overview

This report documents the Week 4 capstone project completed as part of the Data Analyst Internship. The work uses a sample sales dataset to clean business records, explore sales and profit patterns, analyse results in Excel, inspect the Power BI dashboard structure and evaluate saved sales predictions.

700	118.73M	16.89M	1,125,806
Sales records	Total sales	Total profit	Units sold

Project objectives

• Prepare the dataset by checking missing values, duplicates, column names and date formatting.

• Explore sales and profit by country, product and month using charts.

• Summarise business performance in Excel using PivotTables and charts.

• Use the saved actual and predicted sales values to evaluate prediction performance.

• Present key business measures and comparisons in an interactive Power BI report.

Data coverage: 01 Sep 2013 to 01 Dec 2014; 5 countries; 6 products; 16 columns.

1. Introduction

Data analysis helps organisations understand past performance, identify patterns and make informed business decisions. In this project, sales data was analysed using Python, Excel and Power BI. Python was used for data preparation, exploratory analysis and prediction evaluation. Excel was used to summarise sales through PivotTables and charts. Power BI was used to build an interactive dashboard for presenting important business measures.

The main aim was to understand how sales and profit differ across countries and products, observe monthly sales behaviour and assess how closely a regression model's predicted sales match actual sales.

2. Dataset Description

The uploaded cleaned workbook contains 700 records and 16 columns. The columns describe customer segment, country, product, discount band, units sold, prices, gross sales, discounts, sales, cost of goods sold (COGS), profit and date-related fields.

Dataset detail	Observed value
Total records	700
Total columns	16
Countries	5
Products	6
Segments	5
Date range	Sep 2013 – Dec 2014
Missing values after cleaning	0 across all 16 columns

Dataset detail	Observed value
Duplicate rows in original notebook check	0

The dataset contains records for Canada, France, Germany, Mexico and the United States of America. The six products are Amarilla, Carretera, Montana, Paseo, Velo and VTT.

3. Data Cleaning

Data cleaning was performed in Python with the Pandas library. The notebook showed 53 missing values in the Discount Band column before cleaning. No missing values were found in the other columns, and the duplicate-row check returned zero.

Cleaning activities completed

• Removed extra spaces from column names using strip().

• Filled missing Discount Band values with the category “No Discount”.

• Checked and removed duplicate rows; the initial duplicate count was zero.

• Converted the Date column to a datetime format.

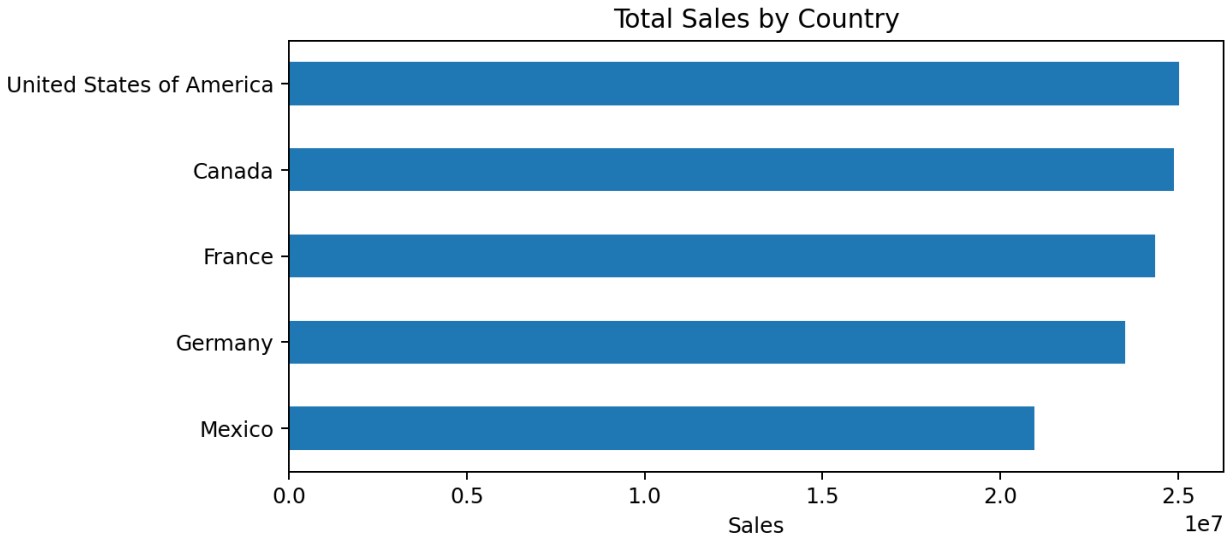
• Checked missing values again; all 16 columns had zero missing values after cleaning.

The cleaned data was exported to an Excel workbook named Cleaned_Sales_Data.xlsx. The cleaned workbook supplied for this report contains the 700-row dataset.

4. Exploratory Data Analysis (EDA)

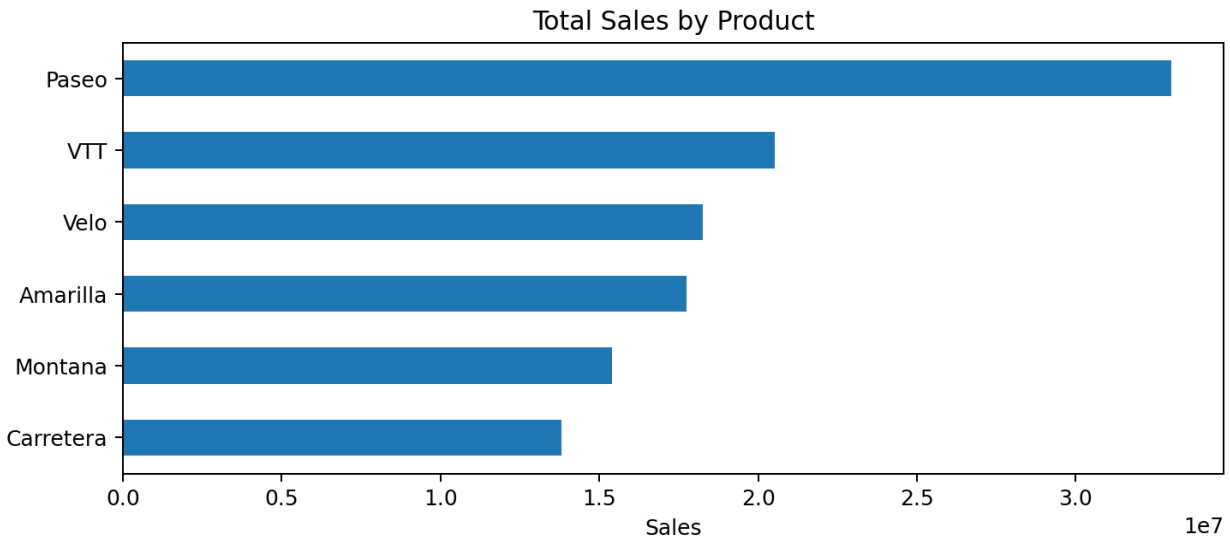
Exploratory Data Analysis was carried out to understand the distribution of sales and profit and to identify differences across countries, products and months. The uploaded notebook includes sales by country, sales by product, profit by country, monthly sales, a profit histogram and a numerical correlation heatmap.

4.1 Sales by Country



The United States of America recorded the highest total sales at 25,029,830.16. Mexico recorded the lowest total sales at 20,949,352.11 among the five countries.

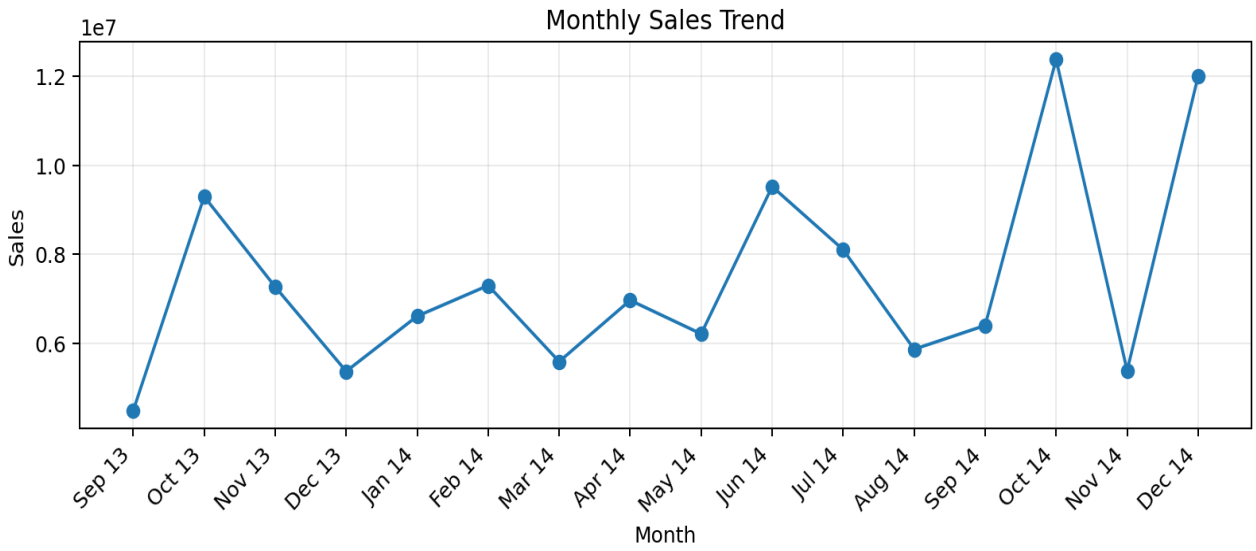
4.2 Sales by Product



Paseo was the highest-selling product by total sales (33,011,143.95), while Carretera recorded the lowest total sales (13,815,307.88).

4. Exploratory Data Analysis (continued)

4.3 Monthly Sales Trend



At the month level across the dataset, October 2014 had the highest recorded sales (12,375,819.92). September 2013 had the lowest recorded sales (4,484,000.03). This is a month-by-month view of the supplied records, not a seasonally adjusted forecast.

4.4 Profit by Country

Country	Total profit
France	3,781,020.78
Germany	3,680,388.82
Canada	3,529,228.89
United States of America	2,995,540.67
Mexico	2,907,523.11

France recorded the highest total profit (3,781,020.78) and Mexico recorded the lowest (2,907,523.11) among the countries in the dataset.

4.5 Other EDA visuals

• Profit Distribution: a histogram of the Profit column was created to view how profit values are distributed.

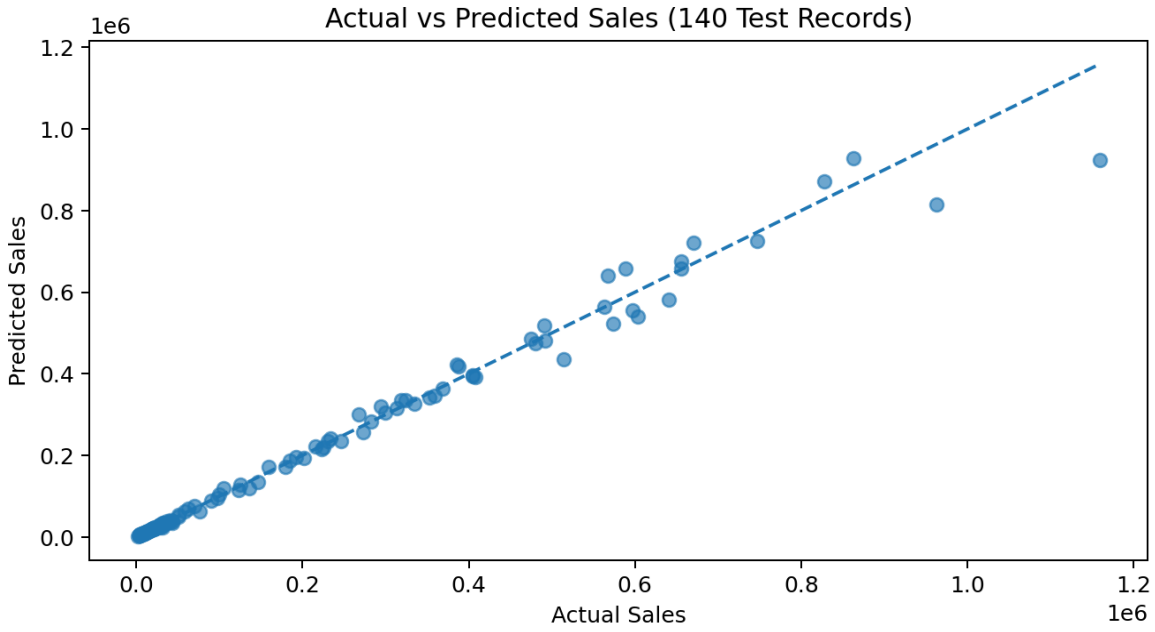
• Correlation Heatmap: a heatmap was created using correlations between numeric columns to explore relationships among numeric measures.

The uploaded notebook contains the code for the profit distribution and correlation heatmap. The report summarises their purpose rather than assigning specific patterns that were not printed as numeric results in the notebook.

5. Regression Model and Sales Prediction

The project brief identifies the prediction approach as Random Forest Regression. The supplied prediction-results workbook contains 140 pairs of Actual Sales and Predicted Sales. The following evaluation scores were calculated directly from those saved pairs. The model-training code and train/test split settings were not present in the supplied results workbook, so those implementation details are not independently verified here.

Evaluation measure	Calculated result	Meaning
Test prediction rows	140	Number of actual/predicted pairs supplied
MAE	11,227.56	Average absolute prediction error
RMSE	29,695.79	Error measure that gives larger errors more weight
R^2 score	0.9841	Proportion of variation explained by the predictions



The points closer to the dashed diagonal line represent predictions closer to the actual sales values. The calculated R^2 score is approximately 0.9841 for the supplied 140 prediction pairs. These scores describe this saved test-output file and should not be treated as a guarantee of future prediction accuracy.

Model workflow described in the project brief

• Select relevant input features and the target variable (Sales).

• Split the dataset into training and testing data.

• Train the Random Forest Regression model on the training data.

• Generate predicted sales values for the test data.

• Compare actual and predicted values and evaluate the results using MAE and R^2 (RMSE is also reported here).

6. Excel Analysis

Excel was used to summarise and compare business sales. The uploaded Week4_Excel_Analysis.xlsx workbook contains a cleaned 700-record data sheet and PivotTable-style summaries for country-wise sales and product-wise sales.

Activities completed

• Created a country-wise sales summary using a PivotTable.

• Created a product-wise sales summary using a PivotTable.

• Compared total sales across five countries.

• Compared total sales across six products.

• Used the summaries to support chart-based interpretation.

Country	Sales
United States of America	25,029,830.16
Canada	24,887,654.89
France	24,354,172.28
Germany	23,505,340.82
Mexico	20,949,352.11

The Excel workbook supplied includes the summary tables. If screenshots of the Excel charts are required by the internship evaluator, they can be inserted alongside this section in a revised copy of the report.

7. Power BI Dashboard

The uploaded Power BI project file, WEEK4_Sales_Dashboard.pbix, was inspected at the report-structure level. It contains 15 report visuals, including four card visuals and four slicer visuals, along with chart visuals. This confirms the presence of dashboard components in the Power BI file; the PBIX file itself does not provide a ready-to-insert screenshot in this report.

Dashboard measures and views listed in the project brief

• Key performance indicators: Total Sales, Total Profit, Total Units Sold and Total Transactions.

• Sales by Country and Sales by Product.

• Monthly Sales Trend and Profit by Country.

• Interactive filters/slicers for Year, Country and Segment.

Dashboard component	Purpose
KPI cards	Show headline business performance measures
Country and product charts	Compare performance across markets and products
Monthly trend	Show how sales change over time
Profit by country	Compare contribution to business profit
Slicers	Filter the report by selected categories

Evidence note: Insert a screenshot of the finished dashboard from Power BI Desktop before submitting if your internship instructions specifically require a dashboard image. The editable .pbix file is a separate project deliverable.

8. Findings and Recommendations

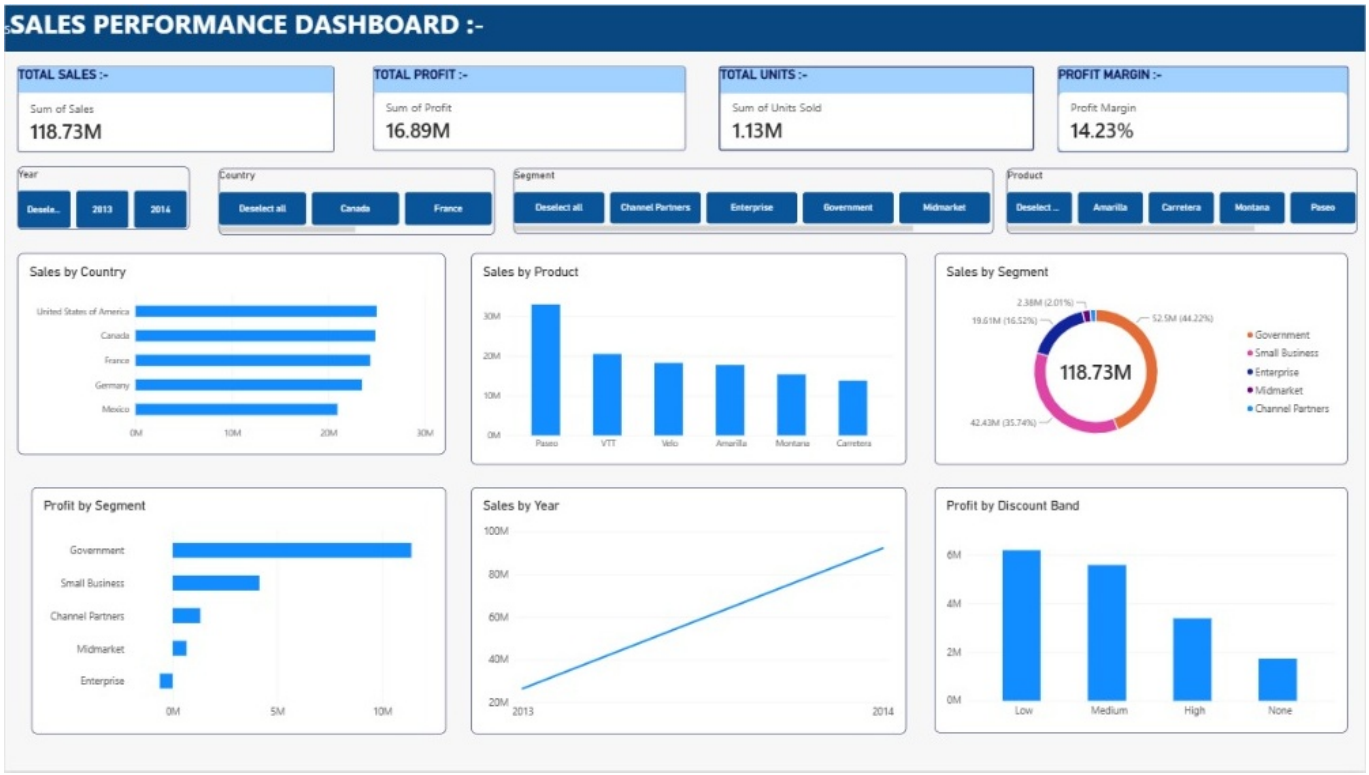
Key findings

• The dataset contains 700 records across 5 countries and 6 products.

• Total sales are 118,726,350.26, total profit is 16,893,702.26, and total units sold are 1,125,806.

Power BI Dashboard – Screenshot Evidence

The screenshot below was supplied for the Week 4 capstone report. It shows the Sales Performance Dashboard with KPI cards, slicers and charts.



• The United States of America has the highest country-wise sales (25,029,830.16).

• Paseo has the highest product-wise sales (33,011,143.95).

• France has the highest country-wise profit (3,781,020.78).

• October 2014 is the highest-sales month in the monthly aggregation.

• The saved regression prediction results produce an R^2 score of 0.9841 across 140 test rows.

Recommendations

• Review the strong sales performance of Paseo and understand which customer segments and markets contribute most to it.

• Investigate the lower sales totals for Carretera and Mexico before deciding whether pricing, promotion or distribution changes are appropriate.

• Monitor country-wise sales and profit together; high sales do not automatically mean the highest profit.

• Use monthly sales patterns to plan inventory and promotional activity, while checking additional years of data before assuming a recurring seasonal pattern.

• Use model predictions as decision-support estimates and monitor prediction errors as new sales data becomes available.

• Review discount levels alongside profit to ensure that sales growth does not reduce profitability.

9. Conclusion

This capstone project provided practical experience in preparing, analysing and presenting business sales data. Python and Pandas were used to inspect and clean the dataset, and charts were created to study country-wise, product-wise and monthly sales patterns. Excel PivotTable summaries supported comparisons, while the Power BI project file contains KPI cards, charts and slicers for interactive reporting. Saved actual and predicted sales values were also evaluated using MAE, RMSE and R^2.

Overall, the project demonstrated how data analysis can turn raw sales records into useful business information. It strengthened practical understanding of data cleaning, exploratory data analysis, spreadsheet reporting, dashboard development and regression evaluation.

Tools and project files

• Python: Pandas, NumPy, Matplotlib and Seaborn (as shown in the notebook).

• Microsoft Excel: cleaned sales data and PivotTable summaries.

• Microsoft Power BI: WEEK4_Sales_Dashboard.pbix.

• Regression output: Regression_model_for_Sales_Prediction_Results.xlsx.

Submitted by: **Janhavi Amol Chavan**

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